AGRI TECH INNOVATION

OFFLINE-FIRST AI

Smart Farming Reimagined.

Empowering Smallholder Farmers with an AI Agronomist in Their Pocket — Operating Seamlessly Anywhere, Online or Offline.

35-40%+
YIELD INCREASE

90%
OFFLINE CAPABILITY

1:3,000
EXTENSION GAP BRIDGED

The Smallholder Challenge

CORE PROBLEM

300M+ Farmers Lack Timely Agronomic Advice

- **Scarcity of Extension Officers:** Extension agent ratios often exceed 1:3,000 farmers, leaving daily crop management to guesswork.
- **Climate Predictability Gap:** Extreme volatility, unexpected pest outbreaks, and soil degradation endanger yields annually.
- **Connectivity Barriers:** Standard agtech apps demand 4G/5G data, completely ignoring rural field realities.



Expert Agronomist in Every Pocket

01

Offline-First Design

Full core functionality operates without internet. Daily insights, field logs, and advisory sync automatically when a 2G/3G link is detected.

02

Conversational AI

An intelligent, multi-LLM agronomic advisor delivering actionable advice in local languages tailored to specific crops and soil conditions.

03

Guided Seasons

Step-by-step workflow tracking crops from intake and planting to daily micro-actions and final harvest yield measurement.

Smart Advisory Features

FEATURE SET A

Daily Intelligence & Advisory

Today Dashboard: Instant snapshots of micro-weather, soil moisture, frost alerts, and high-priority farm tasks.

AI Assistant: Multi-turn agronomy chats supporting text & leaf photo uploads with prioritized urgency badges.

REAL-TIME MICRO-CLIMATE ENGINE

FEATURE SET B

Field & Farm Operations

Boundary Mapping: Draw polygon field boundaries on satellite maps to compute precise acreage and monitor soil health.

IoT Telemetry: Seamless integration with on-farm soil moisture probes and local weather stations.

PRECISION SATELLITE POLYGONS

Guided Farm Season Workflow

STEP 01

Setup & Map

Draw field boundaries on satellite maps and complete quick crop intake questionnaires.

STEP 02

Stage Tracking

Automated growth stage tracking (Planning → Planted → Growing → Harvest).

STEP 03

Daily Actions

Receive localized daily alerts for irrigation, fertilizing, and pest mitigation.

STEP 04

Log Yields

Record harvest metrics to continuously refine location-specific AI advice.

AI & On-Device Vision

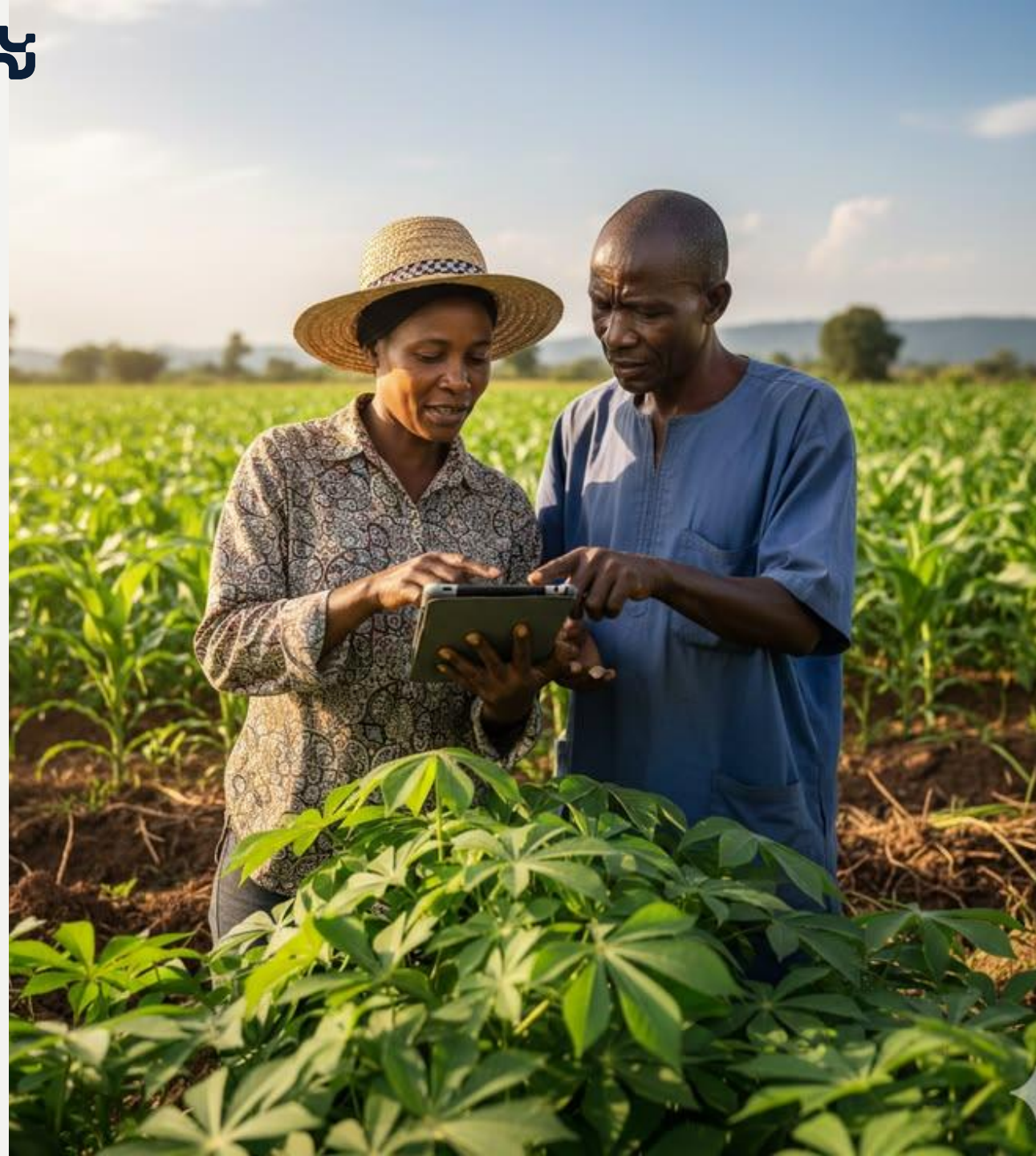
HYBRID ENGINE

Cloud & Client Intelligence

Combines server-side LLMs (GPT-4o & Claude 3.5) with local, on-device AI model execution.

On-Device Leaf Diagnostics

TensorFlow.js and MobileNet classify crop leaf diseases directly on cheap mobile devices—zero data consumed, instant diagnosis for farmers.



Low-Bandwidth Architecture

- **Progressive Web App (PWA):** Fully installable application shell powered by Workbox service workers and ultra-lightweight JSON payloads (<15KB).
- **IndexedDB / Dexie Offline Storage:** Local-first database manages field logs, offline diary notes, and chats, queuing syncs until connectivity returns.
- **Django 5 & DRF Backend:** High-performance Python backend orchestrating Open-Meteo weather integrations and automated CRON alert jobs.
- **VAPID Web Push Notifications:** Proactive morning alerts delivered directly to device notification bars over low-grade 2G/3G networks.

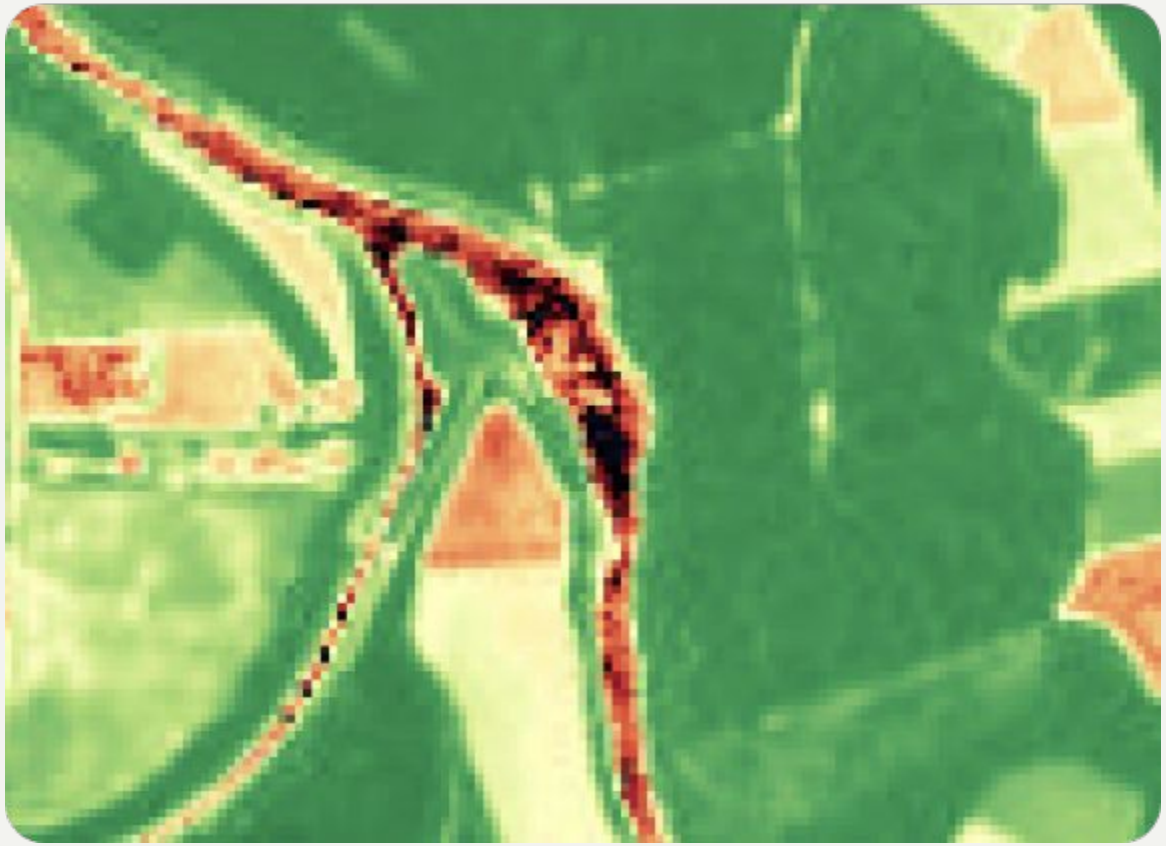
Field Mapping & IoT Sensors

PRECISION AG

Mapping Smallholder Plots

Farmers map field plots down to fractional hectares using interactive satellite polygon tools.

- **Color-Coded Plots:** Categorize fields by crop type and stage for crop rotation.
- **IoT Telemetry Integration:** Real-time soil moisture and environmental weather station charting via lazy-loaded charts.
- **Hyper-Local Weather:** Integrates Open-Meteo point-location soil and micro-climate data.



Transforming Rural Yields

35%+

TARGET YIELD INCREASE

Substantial ROI for Smallholders

By shifting farming from intuition-based decisions to climate-aware precision guidance, Asili drastically cuts crop losses due to undetected frost, drought, and pest outbreaks.

EXTENSION MULTIPLIER

Empowers local Development Agents (DAs) and agricultural cooperatives to support 10× more smallholder farmers efficiently.

Strategic Growth Roadmap

PHASE 01 — ETHIOPIA PILOT

Deployment across key farming regions (Teff, Maize, Wheat, Coffee in different seasons).

PHASE 02 — LOCAL LANGUAGES

Full localization for Amharic, Afan Oromo, and Tigrinya with voice interaction.

PHASE 03 — COOPERATIVE PORTAL

Analytics dashboard for agricultural extension services and regional cooperatives.

PHASE 04 — EAST AFRICA SCALE

Expanding across Kenya, Uganda, and Rwanda with regional crop calendar datasets.

PARTNER WITH ASILI

Building Resilient Climate Agriculture.

Join us in delivering next-generation offline AI agronomy to millions of smallholders.